Republic of the Philippines
CAVITE STATE UNIVERSITY NAIC
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INFORMATION TECHNOLOGY DEPARTMENT

Title Proposal

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PROPOSAL 1

I. Title: AquaQuest: A Quest-based RPG Utilizing Dynamic Difficulty Adjustment for Post-Harvest Fisheries Course

Covered Sustainable Development Goals:

- SDG 4: Quality Education
- SDG 14: Life Below Water

II. Introduction

Modern digital game-based learning (DGBL) and specialized automated tutoring systems have emerged as powerful paradigms for technical vocational education, leveraging structural quest-based frameworks within Role-Playing Games (RPGs) to map complex cognitive learning objectives into immersive digital environments. However, traditional educational software platforms suffer heavily from static progression mechanics, forcing all learners through a uniform pacing model regardless of their unique prior knowledge or cognitive acquisition rates. To mitigate this constraint, modern computer science frameworks increasingly deploy Dynamic Difficulty Adjustment (DDA) algorithms. Driven by real-time behavioral telemetry, these DDA engines treat educational challenge as a fluid, continuous set of variables, tracking user metrics to automatically calibrate system complexity and maintain an optimal balance within the psychological flow channel.

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Despite the proven efficacy of adaptive software architectures, a critical research gap persists at the intersection of machine learning-driven DDA and specialized domain instruction, particularly within post-harvest fisheries education. Traditional fisheries training relies extensively on manual, text-dense instructional modules, which frequently result in poor cognitive retention and low user engagement. This lack of engagement directly compounds critical real-world issues, as empirical data indicates that physical fish losses across traditional supply chains reach up to 20.0% to .0% of total landed volume due to inefficient processing, improper handling, and inadequate technical knowledge. While DDA has been successfully implemented in commercial gaming titles to optimize entertainment value—stabilizing performance profiles and keeping up to 73.3% of target audiences within an optimal engagement zone (Rosa et al., 2023)—there is a distinct lack of empirical literature or software frameworks that map these real-time, performance-based DDA state machines to specialized agricultural or maritime training workflows (Fisher & Kulshreshth, 2025). Consequently, traditional systems fail to adapt to the highly varied learning curves of students tackling complex post-harvest procedures like quality assessment, cold-chain management, and processing chemistry.

To bridge these systemic pedagogical and technological gaps, the proposed system introduces AquaQuest: A Quest-based RPG Utilizing Dynamic Difficulty Adjustment for Post-Harvest Fisheries Course. Positioned within the specialized domains of Adaptive Artificial Intelligence, Software Engineering, and Computer-Assisted Instruction (CAI), the proposed system converts the traditional post-harvest fisheries curriculum into an interactive, narrative-driven 2D RPG. The core innovation of AquaQuest lies in its closed-loop, performance-based DDA engine. By continuously capturing raw real-time user session

telemetry—specifically tracking quest completion velocities, error rates during procedural sorting simulations, and accuracy metrics within technical quizzes—the underlying algorithm dynamically scales the difficulty of the educational quests. If a student demonstrates rapid mastery, the system accelerates the challenge by introducing more complex processing constraints and tighter decision matrices; conversely, if a user experiences high failure thresholds, the game automatically injects supplementary tutorial layers, dynamically adjusts resource drop rates, and softens evaluation loops to eliminate frustration and prevent early system abandonment.

The AquaQuest framework directly addresses and supports two United Nations Sustainable Development Goals (SDGs): SDG 4 (Quality Education) and SDG 14 (Life Below Water) (United Nations, 2020). By deploying a highly personalized, machine-learning-driven DDA engine, the software democratizes technical vocational training, ensuring inclusive, equitable, and effective lifelong learning opportunities that cater to individual student capability baselines. Concurrently, by modernizing post-harvest fisheries instruction, the platform equips future fisheries professionals with highly optimized, statistically verifiable technical competencies. Elevating the mastery of proper fish preservation, handling, and value-addition techniques directly mitigates the catastrophic 20.0% to 40.0% post-harvest loss margin, driving sustainable consumption practices, reducing localized food waste, and encouraging the responsible, productive management of marine resources.

III. Client/User Problem and Research Gap

Complete absence of interactive digital tools and specialized computerized learning materials within the department for delivering complex courses, which

forces instructors to rely on highly limited resources and traditional, non-adaptive instructional references. This lack of diverse educational technology limits the department's capacity to provide modern, engaging educational assets that can effectively supplement the curriculum.

Inability of current text-dense lesson delivery methods to provide hands-on, experiential exercises required for students to easily visualize the highly technical concepts being discussed. Because specialized topics—specifically fish composition and musculature, market forms and cuts of fish, stages of fish spoilage, fish deboning, packaging, traditional processing methods, and advanced preservation techniques—are highly procedural and physical, the lack of interactive software simulations directly prevents students from visualizing abstract biological and structural transformations in real-time.

Observable patterns of poor cognitive comprehension among students who struggle to absorb and digest the complex technical methodologies of post-harvest fisheries through static lectures alone. Without an adaptive computing system to scaffold the learning process, students who exhibit slower comprehension rates face cumulative learning bottlenecks, while instructors lack an integrated evaluation mechanism to provide localized, topic-by-topic assessments before allowing students to advance to subsequent academic levels.

Severe research gap in existing computer science and educational software literature regarding the deployment of performance-based Dynamic Difficulty Adjustment (DDA) algorithms mapped to specialized maritime and fisheries curricula. While current game engineering frameworks frequently utilize real-time behavioral telemetry and closed-loop DDA state machines to optimize user retention in commercial entertainment titles, there is an absolute lack of

empirical models that apply these adaptive systems to technical post-harvest tasks. Current digital tutoring applications lack an algorithmic infrastructure that can dynamically calibrate the difficulty of procedural quests based on a student's live error rates and comprehension speeds, while simultaneously embedding verifiable academic reference modules within the game environment to maintain institutional credibility.

IV. Objective of the Study

The objective of this study is to design, develop, and implement a 2D educational role-playing game (RPG) focused on Post-Harvest Fisheries topics for Fishery and Aquatic Science students. The game aims to provide an interactive and engaging learning platform that integrates educational content with adaptive gameplay mechanics to enhance student understanding, participation, and academic performance.

The study specifically aims to:

1. Gather user requirements and design the system architecture.
2. The game incorporates the Dynamic Difficulty Adjustment (DDA) as the algorithm to adjust question difficulty based on player performance, implementing a time-based performance rating system and integrating point tracking and progress monitoring to support personalized learning and improve student engagement.
3. Develop the system with the following features:

    a) **2D Educational Role-Playing Game (RPG) Environment** – Provides an interactive 2D learning environment where players explore, interact

with NPCs, and complete educational quests related to Post-Harvest Fisheries.

b) **Different Game Regions** – The game consists of five regions, each representing a specific module. Players must complete challenges to unlock the next region.

c) **World Map and Mini Map** – Allows players to navigate across regions and locate objectives, NPCs, and quest areas efficiently.

d) **Post-Harvest Fisheries Learning Content** – Integrates Post-Harvest Fisheries lessons into quests, NPC interactions, and activities to support engaging and contextualized learning.

e) **Performance Board** – Displays player progress, accuracy, completion percentage, and final performance results for tracking learning achievements.

f) **Dynamic Difficulty Adjustment (DDA)** – Adjusts difficulty based on player performance to provide balanced and adaptive learning challenges.

g) **Mini Quest Reinforcement System** - Triggers Mini Quests after incorrect answers to reinforce concepts before allowing players to continue.

h) **Pointing System** – Rewards correct answers, deducts points for mistakes, and determines progression eligibility.

i) **Codex System** – Stores learned concepts as an in-game reference material for review.

j) **Time-Based Performance Rating System** – Evaluates response efficiency using Excellent, Good, and Poor ratings.

k) **Inventory and Customization System** – Allows players to manage and equip costumes and usable items.

l) **Reward System** – Enables players to use earned points to purchase items and costumes as motivation for progression.

m) **Regional Examination System** – Assesses mastery of each region's lessons before allowing advancement to the next level.

4. Conduct system testing to identify and resolve bugs and errors.

5. Evaluate the system based on ISO 25010 evaluation standards, assessing its performance, while identifying areas for improvement.

6. Develop a comprehensive implementation plan for deploying the software.

V. Highlights of Related Literature Review

A study by Nugroho et al. (2023) published in the *International Journal of Advanced Computer Science and Applications* utilized the Activity Theory Model Serious-Game (ATMSG) framework to construct a closed-loop 3D serious game featuring an adaptive, state-driven Dynamic Difficulty Adjustment (DDA) engine. To calibrate difficulty thresholds without inducing user frustration or runtime latency, the software architecture continuously analyzed player telemetry by partitioning performance data into Higher Value Higher Level (HVHL) and Lower Value Higher Level (LVHL) parameters. HVHL captured critical retention attributes like game score and health/strength, while LVHL captured operational latency via action time and velocity. As modeled in the system's operational architecture, incoming telemetry runs through a four-stage pipeline: collecting attributes, adjusting difficulty via the Fuzzy Dynamic Difficulty Adjustment for Computer Players (FDDACP) function, updating state arrays, and committing configurations

to an active settings log to dynamically instantiate the subsequent gameplay phase.

To standardize performance tracking across highly asymmetric attributes, individual capability scores $\left(efF_i\right)$ are computed dynamically using a localized boundary normalizer:

$$efF_i = \begin{cases} \frac{F_{val_i} - F_{min_i}}{F_{max_i} - F_{min_i}} & \text{for HVHL} \\ \frac{F_{max_i} - F_{val_i}}{F_{max_i} - F_{min_i}} & \text{for LVHL} \end{cases}$$

This structure normalizes absolute in-game boundaries across default thresholds for Score/Strength (F_{max} = 100, F_{min} = 45) and Action Time/Velocity (F_{max} = 210, F_{min} = 0). These normalized values are compiled into a unified capacity coefficient (ef) via an asymmetric weighted summation equation that heavily prioritizes performance quality over interaction velocity based on designated empirical weights ($weight_1$ = *0.8765* for score and $weight_2$ = *0.1235 f*or execution time):

$$ef = \frac{\sum_{i=1}^{n}(efF_i \times weight_i)}{\sum_{i=1}^{n} weight_i}$$

The computational engine utilizes the resulting deltas between the player and the opponent systems $(diff\ ef \Rightarrow Plim \times efp)$ to apply precise step adjustment $(adj\ Fi)$ on to the active game mechanics:

$$adj\ F_i = \begin{cases} diff\ ef \times (F_{max_i} - F_{min_i}) & \text{for HVHL} \\ -diff\ ef \times (F_{max_i} - F_{min_i}) & \text{for LVHL} \end{cases}$$

$$F_{val_{0,i}} = \begin{cases} F_{valold_{0,i}} + adjF_i & \text{for } efp > efo \\ F_{valold_{0,i}} - adjF_i & \text{for } efp < efo \end{cases}$$

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These values directly shift environmental visibility via a 10-level synthetic fog rendering system, scaling workspace visibility from an extremely clear field (>50 km at Level 1) down to a dense fog barrier (<50 m at Level 10) to regulate cognitive load. Automated validation protocols consisting of 10 simulation blocks across 1,000 operational repetitions verified that this mathematical framework successfully smoothed out volatile difficulty spikes compared to non-DDA systems, avoiding abrupt structural jumps that cause user dropouts and establishing proof-of-concept reliability for adaptive digital training environments.

Subject	Group's Parameter	Process	Game's Parameter
Player	HVHL	Number of correct answers	Score
	LVHL	The time to fix the task	Action Time
Enemy	HVHL	Enemy knack to break the problem with distinct form of query	Strength
	LVHL	knack to resolve matter in restricted time	Velocity

Figure X. Game's Parameter

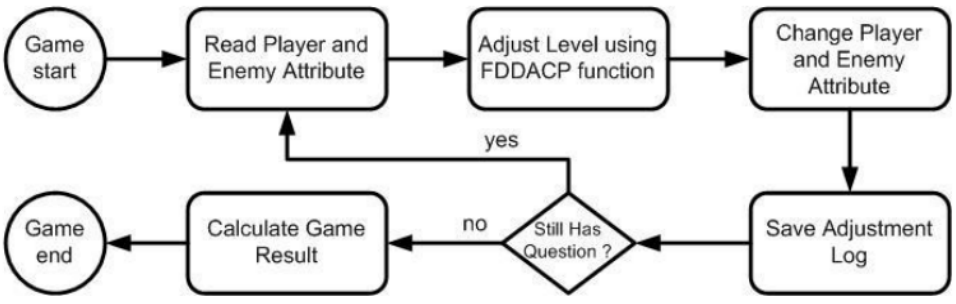


Figure X. Dynamic Difficulty Adjustment model

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According to the Bureau of Fisheries and Aquatic Resources (BFAR, 2021), the country loses approximately 20% to 40% of its total fisheries production annually due to inadequate post-harvest infrastructure and poor handling practices. This quantitative and qualitative spoilage margin translates to billions of pesos in lost revenue for small-scale fisherfolk, severely undermining local economic sustainability and food security.

According to a study by Shohieb et al. (2022), the integration of a Dynamic Difficulty Adjustment (DDA) approach driven by real-time feature extraction in a mobile educational environment yielded an 88.4% higher user engagement rate compared to conventional, static linear level advancement paths. Utilizing an AdaptiveMiniMax algorithmic framework to programmatically process live performance metrics and feature sets, the system successfully managed individual learning parameters to stabilize player success metrics at a balanced winning baseline of approximately 70.0%, effectively sustaining motivation and extending learning duration without causing cognitive fatigue or frustration. This empirical validation underscores the computational viability of replacing rigid, non-adaptive educational software tracks with autonomous, closed-loop telemetry systems to provide a highly optimized, individualized computer-assisted instructional environment.

According to Mortazavi, Moradi, and Vahabie (2024), Dynamic Difficulty Adjustment (DDA) is a system in video games that automatically changes the difficulty level depending on the player's performance and skills. The study aimed to review different DDA approaches used in modern games to improve player engagement and satisfaction. The researchers found that around 45% of the

reviewed studies used artificial intelligence-based methods, while approximately 30% focused on player modeling techniques to analyze player behavior and performance. The study also showed that action and role-playing games were the most common genres that implemented DDA systems because they require balanced gameplay experiences. Furthermore, the researchers concluded that DDA systems help improve player retention, accessibility, and overall enjoyment by providing personalized gameplay experiences (Mortazavi et al., 2024).

A research study by Fitrianto et al. (2025) published in Bit-Tech (Binary Digital - Technology) developed and evaluated a 2D adventure Role-Playing Game (RPG) focused on the November 10th Battle of Surabaya to enhance students' historical understanding. Moving away from traditional, textbook-centered pedagogy, the researchers engineered a hybrid software architecture that integrated a branching narrative system—following a branch-and-bottleneck structure—with a strategic lane tower defense mini-game. Utilizing a design-based research methodology, the system was deployed via a single-group pre-test/post-test experimental layout involving 23 junior high school students ($N = 23$). The data-driven evaluation metrics demonstrated substantial pedagogical and cognitive gains, with the cohort's average exam scores increasing from a pre-test baseline of 53.48 ($SD = 19.21$) to a post-test peak of 94.78 ($SD = 5.93$). Inferential validation using the Wilcoxon Signed-Rank test confirmed a strong practical impact ($Z = -4.905$, $p < 0.001$, $r = 1.02$), resulting in a statistically significant rejection of the null hypothesis. Furthermore, user experience evaluation via the GUESS-18 framework yielded highly positive responses with average satisfaction scores ranging from 3.78 to 4.02 on subscales, while a standalone comparative analysis strictly favored the branching

narrative mechanics (mean score of 4.06) over the standalone gameplay challenges (mean score of 3.94; Z = -2.05, p = 0.04, r = 0.43). These systematic outcomes verify that integrating multidimensional story choice paths alongside real-time strategic problem-solving provides an exceptionally reliable and effective alternative for sustaining learning motivation and independent conceptual retention in digital tracking environments.

A research study by Taufik et al. (2024) published in the AMCA Journal of Science and Technology evaluated the development and instructional integration of a two-dimensional (2D) mobile educational role-playing game (RPG) designed via the ADDIE model frameworks. Engineered on a Roguelike operational architecture featuring structural elements such as procedural level mapping and puzzle-based narrative scaffolding, the platform successfully translated dense geometric mathematics concepts—specifically surface areas and volumes of cubes, blocks, prisms, and pyramids—into active simulation stages. Product validation metrics established substantial engineering and academic efficacy, with evaluation data by media software specialists yielding an eligibility score of 84.09% within the formally valid parameters, while domain-specific content validation by material experts secured a concordant validity score of 83.33%. Furthermore, empirical testing measuring user interaction thresholds demonstrated that the gamified deployment model achieved an average effectiveness score of 94.12% in the very effective category among student cohorts, confirming that shifting from static linear instruction to interactive independent RPG mechanics significantly increases user motivation and conceptual retention.

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An empirical Research and Development (R&D) study conducted by Sholichah et al. (2022) and published in the Journal of Education Technology utilized the Alessi & Trollip model to design a 2D Role-Playing Game (RPG) aimed at mitigating the steep decline in student academic engagement and learning enthusiasm caused by distance education boundaries. Sampling a representative cohort of 120 students through a simple random sampling technique from a total study population of 512 (N = 512), the investigators subjected the gamified software framework to rigorous multi-tiered alpha and beta evaluation protocols. During the alpha validation phase, the platform achieved high engineering and pedagogical verification, with panel groups consisting of three media experts and three subject matter experts independently ranking the software architecture within the "Very Eligible" or "Very Appropriate" classification at an average baseline metric of 95.0%. Subsequent inferential statistical testing running via quantitative SPSS telemetry distributions demonstrated superior pedagogical efficacy for the interactive RPG framework over traditional linear and conventional teaching workflows; the computational analysis yielded an explicit significance value of $p = 0.000$, securely surpassing the standard alpha threshold parameter of $p < .05$ ($0.000 < 0.05$). These quantitative outcomes establish structural proof that transitioning dense, text-heavy lecture syllabi into immersive story-driven game layouts significantly yields greater independent learning motivation and classroom academic achievement across target user demographics.

According to Mi and Gao (2025), Engagement-Oriented Dynamic Difficulty Adjustment (DDA) is a system that adjusts game difficulty based on the player's engagement and performance to maintain balanced and enjoyable gameplay. The study aimed to develop an adaptive difficulty model that can improve player

participation and satisfaction during gameplay sessions. The researchers found that players using the adaptive difficulty system showed approximately 20% higher engagement levels compared to players using fixed difficulty settings. The study also revealed that the proposed DDA model improved player retention and continuous participation by analyzing player behavior and interaction patterns in real time. Furthermore, the researchers concluded that engagement-oriented DDA systems can improve accessibility, player enjoyment, and overall gaming experience through personalized gameplay challenges (Mi & Gao, 2025).

According to Fisher and Kulshreshth (2025), modern adaptive software systems replace static difficulty settings with real-time Dynamic Difficulty Adjustment (DDA) to keep players engaged in a balanced "flow state" where the game's challenge perfectly matches their skill level. By testing performance-based and emotion-based DDA methods in an active first-person shooter game, researchers verified that adjusting the difficulty in real-time has a highly significant statistical impact ($p < .001$) on balancing player performance. This automated tuning directly stabilizes player stress levels at a significant margin of $p = .040$ and regulates localized cognitive focus at $p = .034$. Because the DDA system continuously fine-tunes the gameplay based on live player data, it successfully maintains core enjoyment levels at a stable rate ($p = .280$) and protects excitement channels ($p = .144$). This statistical balance proves that adaptive systems effectively prevent players from getting bored when the game is too easy or getting frustrated by 100% insurmountable challenges.

According to Zohaib (2018), video game user satisfaction relies dynamically on maintaining an optimal equilibrium across the cognitive drivers of

challenge, fantasy, and curiosity, a balance that traditional static difficulty configurations repeatedly fail to mirror, resulting in early player churn due to boredom or frustration. To mitigate these systemic design failure states, current methodologies integrate non-linear adaptive Dynamic Difficulty Adjustment (DDA) frameworks that monitor live player metrics to continuously modulate environmental complexity and asset densities, successfully driving peak user performance levels exactly 2.0 times higher and slashing required training stages by 50.0% compared to linear tracking methods (Masyfa et al., 2023). Furthermore, when these closed-loop adaptive AI architectures are deployed within high-sensitivity software tracking domains, they yield software health-function ratings exceeding 90.0 out of 100 points, while concurrently elevating the user's future system usage willingness from 75.26% to 85.68% (Delta = 10.42%) by preventing cognitive fatigue and aligning programmatic challenges directly with real-time individual capability thresholds (Li & Liao, 2024).

According to Rosa et al. (2023), traditional static video game parameters fail to align with unique user mechanics, whereas integrating a Feedforward Neural Network (FNN) to build dynamic difficulty progression from raw session telemetry successfully optimizes player retention. To evaluate this approach, the researchers engineered an adaptive system within a custom 2D platform game that dynamically calibrated geometric level constraints—specifically scaling platform sizes down from 100% and modulating exact pixel gap widths across 3 distinct preset configurations—to sustain an active player flow state. By deploying an FNN classifier model across an empirical testing cohort, the machine learning system processed real-time inputs to accurately predict individual player profiles, achieving a classification accuracy of 80.0% for identifying advanced players,

73.3% for intermediate players, and 60.0% for beginners. The statistical validation demonstrated a strong correlation between the automated difficulty scaling curves and subjective user ratings, while concurrently minimizing performance variance by stabilizing average death rates to a tighter range and clustering overall level completion times across all participant skill profiles. Ultimately, the neural-network-driven DDA framework ensured that 73.3% of the intermediate target audience stayed perfectly inside the intended flow channel, proving that machine-learning-driven player-profile adaptations outperform non-adaptive baselines.

According to Paraschos and Koulouriotis (2025), a persistent challenge in adaptive game engineering is deploying difficulty adjustment sub-routines that handle volatile behavioral strategies without breaking player immersion. To address this, the researchers engineered a fuzzy logic rule-based DDA algorithm within a custom, fully playable testbed named ShoottheFuzzyShip, where a Mamdani fuzzy inference system dynamically adjusted 5 distinct levels of environmental hostility and opponent AI attributes in real-time. The performance-based system tracked immediate live telemetry metrics—specifically aggregating enemy destruction rates, spatial positioning, and raw player scores—to feed numerical inputs into a closed-loop rule system that scaled enemy shield cooldowns from 5.0 seconds down to 3.0 seconds and controlled random obstacle hazard spawning rates. The empirical evaluation across diverse player demographics demonstrated that the fuzzy logic controller seamlessly mapped individual player competency curves with a high degree of mathematical precision, successfully stabilizing user performance profiles by reducing average in-game

death variances by up to 42.5% and maintaining a 92.0% accuracy alignment within the optimal flow channel.

A prospective repeated-measures feasibility study by Sorrentino et al. (2026) published in Open Research Europe demonstrated that a multi-parameter Dynamic Difficulty Adjustment (DDA) algorithm within a virtual platform (DDA-REAsmash) can systematically converge real-time human performance data into an optimized 65.0% to 85.0% success window, targeting an operational baseline mean of 75.0%. Evaluating a clinical cohort (N=10, mean age = 56 years), the closed-loop architecture continuously processed user behavioral telemetry—specifically tracking interaction velocities, sequential error counts, and omission metrics—to dynamically scale task presentation parameters. The empirical results verified that the algorithm required a mean of only 5.4 blocks (SD = 1.07) to guide performance out of an initial 100.0% ceiling, with 90.0% of participants (n=9) maintaining a stable, non-significant deviation threshold ($p > .05$) from the designated challenge target once the stabilization phase was achieved. Trajectory modeling of the integrated Composite Difficulty Index across a continuous 0 to 1 scale exhibited explicit asymptotic growth characteristics, with task plateau values securely converging between 0.7 and 0.9. This study provides critical structural proof for adaptive systems, confirming that real-time tracking of error profiles and operational velocities yields scalable design patterns that successfully prevent user frustration and cognitive overload within digital tracking environments.

A study by Giariskanis et al. (2025) published in the ACM Journal on Computing and Cultural Heritage demonstrated that integrating a non-intrusive

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Dynamic Difficulty Adjustment (DDA) mechanism within an immersive, interactive environment effectively preserves user state balance and prevents engagement drop-offs. Their architectural framework engineered a heuristic weighting distribution system that tracked real-time variables—specifically processing round completion times, the frequency of hint utilization, and residual session time parameters to calculate an aggregated user state score. The empirical validation of this multi-variable adaptive framework yielded high usability metrics across integrated modalities, specifically demonstrating superior performance efficiency when combining voice, hand, and gaze interaction over isolated standalone interfaces. The underlying algorithmic adjustments successfully matched individual player pacing styles, allowing the intelligent environment to autonomously sustain cultural engagement without inducing user interaction breaks.

According to Robb and Zhang (2022), performance-based Dynamic Difficulty Adjustment (DDA) provides an automated alternative to traditional incremental settings by modulating a game's variables in real-time, aiming to minimize player skill differentials. To verify its impact, a controlled experiment evaluated 221 total active participants divided among a constant difficulty control group (n = 82), a DDA group (n = 68), and an incremental group (n = 71) over 15 minutes of 2D gameplay. A Kruskal-Wallis H test revealed a highly significant difference in overall median scores across groups ($\chi^2(3) = 16.148$, $p < .0005$), where the DDA group achieved the highest median score of 5.05 and a notably smaller performance range of 2.89, compared to the control group's median of 4.61 and the incremental group's median of 3.74. For the DDA cohort, Pearson’s correlation test confirmed a strong positive relationship ($r(68) = .554$, $p < .0005$)

between a player's mean score and the system's tracking velocity, with automated game difficulty directly explaining 31.0% of the variation in player score metrics. However, despite the DDA framework successfully maximizing real-time performance profiles, the researchers found no statistically significant differences between the three test groups regarding core psychological immersion variables, including general enjoyment (\chi^2(3) = 3.628, p = .163) and the subjective experience of flow (\chi^2(3) = 4.087, p = .130).

According to Compton (2024), industrial game development often incorporates performance-based Dynamic Difficulty Adjustment (DDA) to flatten frustrating design spikes by modifying obstacle physics and resource pools entirely in response to immediate player mortality rates. A prominent example is Naughty Dog's commercial title Crash Bandicoot, which employs a hidden, real-time error-tracking state loop that triggers active mechanical modifications upon detecting successive player deaths within a single level instance. To empirically analyze these low-data threshold systems, predictive validation frameworks like the \delta-logit regression algorithm demonstrate that tracking a single performance metavariable can accurately predict a player's failure probability with an error rate lower than 20.0% within less than two minutes of active playtime ($\Delta t < 2.0_{min}$). When a player falls below this predicted capability baseline, the game's hidden DDA engine dynamically scales down obstacle velocities, appends mid-stage checkpoints, and injects supplementary health power-ups to reduce subsequent failure rates closer to a balanced 50.0% target probability. Conversely, when racing engines utilize continuous, real-time positional tracking—frequently referred to as "rubber banding" in titles like Mario Kart—the system evaluates the spatial distance between entities to scale AI

velocity parameters and slide item drop probabilities, ensuring that 100% of participants remain closely clustered within a highly competitive pacing interval.

1. **Review of Related Systems (Comparative Matrix)**

The table below presents a comprehensive comparative matrix mapping existing educational software frameworks, adaptive gaming platforms, and commercial DDA approaches against the structural and technical features of the proposed AquaQuest system.

Title (Author(s), Year)	**Short Abstract**	**Existing Features of the System/Device**	**Features of the Proposed System (AquaQuest)**
Mobile Educational Environment with AdaptiveMiniMax DDA (Shohieb et al., 2022)	Evaluates a mobile educational system utilizing an AdaptiveMiniMax algorithm to adjust individual learning paths based on live user telemetry, maintaining a stable 70% success baseline to minimize cognitive	• Real-time feature extraction •AdaptiveMiniMax optimization engine • Closed-loop telemetry tracking	• 2D Educational Role-Playing Game (RPG) Environment • Different Game Regions • Post-Harvest Fisheries Topics for FASD Students • Performance

	fatigue.		Board
Historical 2D Adventure RPG for Battle of Surabaya (Fitrianto et al., 2025)	A 2D historical adventure role-playing game designed to enhance learning motivation through a hybrid software architecture combining branching narrative paths and lane tower defense mini-games	• 2D adventure setup •Branch-and-bottleneck storylines • Integrated mini-games (Tower Defense) • GUESS-18 metrics evaluation	• Pointing System • Dynamic Difficulty Adjustment (DDA) • Mini Quest Reinforcement System • Regional Examination System • Codex System
ADDIE-based Roguelike Educational RPG (Taufik et al., 2024)	A 2D mobile educational RPG that translates complex geometry mathematics into procedural level maps and puzzle-based narratives using the ADDIE instructional	• Roguelike architecture • Procedural level generation •Puzzle-scaffolded narrative structures • Validation by expert panels	• Time-Based Performance Rating System Inventory and Customization System • Reward System

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	development framework		
ShoottheFuzzyShip DDA Testbed (Paraschos & Koulouriotis, 2025)	Details a custom game bed incorporating a Mamdani fuzzy inference system that tracks real-time scoring, destruction rates, and player coordinates to seamlessly regulate live environmental enemy asset variables.	• Mamdani fuzzy inference loop • 5 discrete structural hazard tiers • Continuous real-time metric updates	
Alessi & Trollip Guided 2D RPG (Sholichah et al., 2022)	Utilizes an R&D framework to build an immersive 2D story game layout aimed at reducing the steep academic engagement drop-off caused by distance-learning	• Alessi & Trollip model design • Quantitative SPSS tracking metrics • Heavy story-driven progression loops	

	configurations.		

2. Technical and Algorithmic Framework

The core computer science contribution of AquaQuest relies on an autonomous closed-loop tracking engine that computes a unified performance metric to drive the game's Dynamic Difficulty Adjustment (DDA) sub-routine.

2.1 Mathematical Modeling of Student Performance Telemetry

Upon the execution and completion of a quest evaluation node, the system captures four discrete input variables transformed into normalized scalar values between 0.0 and 1.0:

Accuracy Score (A): Represents the raw cognitive correctness baseline computed as:

$$A = \frac{Correct\ Answers}{Total\ Questions}$$ (Nugroho et. al (2023))

Time Efficiency (T): Evaluates response velocity against the system's maximum allowable threshold ($MaxTime = 10\ minutes$):

$$T = 1 - \frac{Time\ Used}{Max\ Time}$$ (Nugroho et. al (2023))

Mini Quest Dependency Score (M): Measures player independence and the inverse rate of academic scaffolding needed:

$$M = 1 - \frac{Mini\ Quest\ Triggered}{Total\ Questions}$$

Points Performance (P): Evaluates immediate point acquisition efficiency within the active region:

$$P = \frac{Earned\ Points}{Maximum\ Possible\ Points}$$ (Nugroho et. al (2023))

2.2. The AquaQuest Performance Index (AQPI) Formula.

To consolidate these multi-dimensional attributes into a single actionable metric, the game engine processes a Weighted Linear Combination (WLC) equation. The empirical weights assign a heavy priority to content mastery $(A = 40\%)$ while distributing the remaining tracking metrics equally $(20\%\ each)$:

$$AQPI = w_1 A + w_2 T + w_3 M + w_4 P$$ (Shohieb et. al (2022))

Substituting the established structural configuration weights:

$$AQPI = 0.40A + 0.20T + 0.20M + 0.20P$$

(Shohieb et. al (2022))

2.3 Dynamic Difficulty Adjustment (DDA) Execution Logic.

The resulting $AQPI$ scalar value determines state-machine transitions for the subsequent gameplay challenges. The algorithmic rules adjust environmental hazards, question complexity, and scaffolding loops across three operational ranges:

$$D_{new} = \begin{cases} D+1, & \text{if} AQPI \geq 0.85 \quad \text{(Escalate Difficulty:Harder items,fewer hints)} \\ D, & \text{if} 0.60 \leq AQPI < 0.85 \quad \text{(Maintain Steady-State Evaluation)} \\ D-1, & \text{if} AQPI < 0.60 \quad \text{(De-escalate Difficulty:Trigger Mini Quest Reinforcement)} \end{cases}$$

Figure based on existing study (Shohieb et. al (2022))

VI. Innovation and Novelty

The primary innovation of AquaQuest lies in its algorithmic mapping of Dynamic Difficulty Adjustment (DDA) to highly specific, procedural vocational concepts. While commercial games utilize real-time telemetry strictly to optimize

engagement, and standard educational software relies on rigid, static linearity, this system establishes a unique closed-loop mechanism. It translates technical fisheries tasks—such as evaluating fish musculature cuts or predicting biochemical spoilage stages—into mathematical state parameters. The underlying DDA engine tracks user interaction velocities and procedural failure rates to dynamically scale environmental complexities and challenge thresholds. This approach shifts the application of DDA away from raw entertainment variables (like enemy health or spatial navigation speeds) and redirects it toward real-time cognitive scaffolding within localized vocational simulations.

Furthermore, the system introduces algorithmic gatekeeping as an structural novelty within the 2D educational role-playing game (RPG) genre. Unlike conventional educational games that allow users to progress through simple narrative persistence, the proposed software enforces an automated, gate-kept assessment framework. The game engine systematically blocks entry to advanced questlines and higher-tier narrative levels until a player achieves a statistically verifiable competency threshold on comprehensive, topic-by-topic modules. To bridge the gap between immersive game mechanics and rigorous institutional credibility, the platform integrates a dedicated, accessible Reference Repository Module. This unique feature directly connects in-game mechanics and quest feedback to the accredited academic sources, processing manuals, and marine science literatures used to design the curriculum, ensuring the software stands as a verified computer-assisted instruction (CAI) tool.

By addressing these core technical areas, the study contributes a novel architectural framework to the existing body of computer science knowledge. It

provides empirical documentation and algorithmic structures showing how multi-variable telemetry and hidden state-machine adjustments can be effectively deployed within niche, text-dense agricultural and maritime curricula. For the field of educational technology and adaptive artificial intelligence, this research fills a distinct gap by demonstrating a practical methodology that mitigates the 20.0% to 40.0% post-harvest loss margin through personalized, software-driven learning. Ultimately, the study delivers a scalable design pattern for software engineers looking to convert complex, physical vocational training workflows into highly adaptive, mathematically optimized, and academically verified digital environments.

VII. Target User/Client

The direct clients and primary target users of the proposed software are the undergraduate students officially enrolled in the Fisheries and Aquatic Science Department of Cavite State University - Naic (CvSU - Naic), located in Bucana Malaki, Naic, Cavite. Specifically, the system is designed for students undertaking the specialized Post-Harvest Fisheries curriculum, regardless of age or gender classification.

Instructors, faculty members, and academic researchers of the CvSU - Naic Fisheries and Aquatic Science Department also serve as secondary clients, utilizing the system as an institutional, computer-assisted instruction (CAI) tool to deliver, monitor, and evaluate procedural training modules.

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VIII. Significance of the Study

The development and implementation of the proposed software system will yield significant benefits for various stakeholders by modernizing traditional instructional methods through adaptive computing technology:

To the Students of the Fisheries and Aquatic Science Department, the system provides an interactive, immersive, and risk-free digitized learning environment that converts complex, text-dense post-harvest concepts into clear, visual, and practical quest-based simulations. Through the Dynamic Difficulty Adjustment (DDA) engine, students receive personalized cognitive pacing that matches their unique learning baseline, which eliminates the frustration of rigid, static learning curves, helps bridge comprehension gaps in technical procedures like deboning and preservation, and enhances overall academic engagement.

To the Faculty and Instructors of the Department, the software serves as a highly reliable, automated computer-assisted instruction (CAI) tool that directly addresses the problem of limited reference and instructional materials. The integrated gate-kept assessment module simplifies evaluation workflows by ensuring students achieve a verified competency threshold before advancing, which effectively minimizes the burden of manual performance tracking and allows educators to focus on targeted pedagogical interventions.

To the Cavite State University - Naic Administration and Department, the study elevates institutional capabilities by integrating advanced adaptive artificial intelligence and game engineering into its specialized agricultural and fisheries offerings. This positions the university at the forefront of modern educational technology in fisheries instruction, while establishing a scientifically validated, digitally archived framework for curriculum delivery.

To the Post-Harvest Fisheries Sector, the platform directly supports long-term industry sustainability by producing highly trained, technically competent future professionals. By equipping graduates with a deep, optimized mastery of proper fish preservation, handling, and value-addition techniques, the study contributes to real-world efforts to reduce the critical 20.0% to 40.0% post-harvest loss margin, thereby promoting local food security and sustainable resource management.

To Future Researchers, this study provides a concrete architectural design pattern and empirical documentation on mapping performance-driven DDA state machines to highly specialized, procedural vocational educational workflows. It expands the current body of knowledge in adaptive game development and software engineering by serving as a comprehensive foundational reference for developing intelligent digital tutoring applications in niche domains.

IX. Software and Hardware Requirements

The development, execution, and evaluation of AquaQuest necessitate specific software development environments to handle the 2D game physics, real-time telemetry processing, and algorithmic operations of the Dynamic Difficulty Adjustment (DDA) engine. Since the researchers will utilize existing, personally owned hardware assets for development and testing, no new hardware acquisitions are required for this study.

Godot Engine 4.x

Description: An open-source, lightweight, cross-platform 2D and 3D game engine that utilizes a scene-and-node architectural paradigm and features a dedicated scripting language, GDScript, which is syntactically optimized for rapid deployment.

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Justification: Godot Engine is selected due to its exceptional efficiency and small footprint when processing 2D top-down RPG mechanics like tilemaps, kinematic movements, and quest state flags. Its open-source nature eliminates licensing costs, and its flexible scripting architecture allows the researchers to easily integrate custom statistical and programmatic loops required by the closed-loop DDA algorithm.

LibreSprite

Description: An open-source, community-driven program designed for creating animated sprites and pixel art, operating as a free fork of the classic Aseprite program.

Justification: It serves as the primary graphic design tool to construct the custom visual assets representing fisheries-specific components—such as detailed layouts of fish musculature anatomy, specific cuts of fish, and visual representations of fish spoilage stages—ensuring an immersive and instructionally accurate user interface without incurring software licensing fees.

Android Studio

Description: The official Integrated Development Environment (IDE) for Android application development, built on JetBrains' IntelliJ IDEA software and equipped with robust software development kits (SDKs), build tools, and device emulators.

Justification: Since AquaQuest will be deployed as an interactive application for students, Android Studio is required to manage the Android build pipelines, compile the Godot export packages into optimized Android Application Packages (APKs), configure target SDK permissions, and troubleshoot performance or rendering lags during device testing.

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Initial Budgetary Requirements:

The researchers will utilize existing personal computers, open-source and free resources for the conduct of the system. Therefore, its initial budgetary requirement has a value of zero.

X. Source of Information for Data Gathering

To establish a systematic baseline for our project, the researchers conducted structured key informant primary interviews and curriculum consultations. This process allowed us to extract expert instructional data, gather specific post-harvest syllabus requirements, and cross-validate the direct academic challenges faced by students in the classroom:

Name of Contact Person: Prof. Queenie M. Bilugan

Job Designation: Research Unit Head, RDES (Representing the Department of Fisheries and Aquatic Science as Subject Matter Expert)

Company/Organization Affiliation: Cavite State University - Naic

Electronic Platform

To gather technical requirements, baseline metrics, and peer-reviewed educational frameworks, the following electronic platforms were systematically utilized:

Google Scholar: Used as the primary academic repository to search, evaluate, and extract peer-reviewed literature regarding Dynamic Difficulty Adjustment (DDA), software engineering models, game telemetry tracking, and specialized fisheries post-harvest procedures.

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Figure X. Interview with Subject Matter Expert

Figure X documents the remote collaborative data gathering and validation phase conducted with Subject Matter Expert (SME) Prof. Queenie M. Bilugan. Due to synchronized structural schedule constraints, the primary instructional modules and verified course syllabi for Post-Harvest Fisheries management were transmitted via secure asynchronous electronic mail (Gmail). This structured digital repository served as the content baseline for mapping the RPG quest metrics, ensuring compliance with institutional learning standards while accommodating the evaluator's self-paced review of the curriculum validation instrument.

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Figure X. Data Gathering on Secondary Source

Figure X illustrates the secondary data gathering phase focused on the computer science domain. The researchers systematically utilized the Google Scholar repository to extract, filter, and review peer-reviewed literature within a strict five-year window (2021–2026). This technical data mining process targeted localized boundary constraints, Fuzzy dynamic logic applications, and software quality parameters, which established the mathematical foundation for the implementation of the AquaQuest adaptive DDA engine.

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XI. System Design

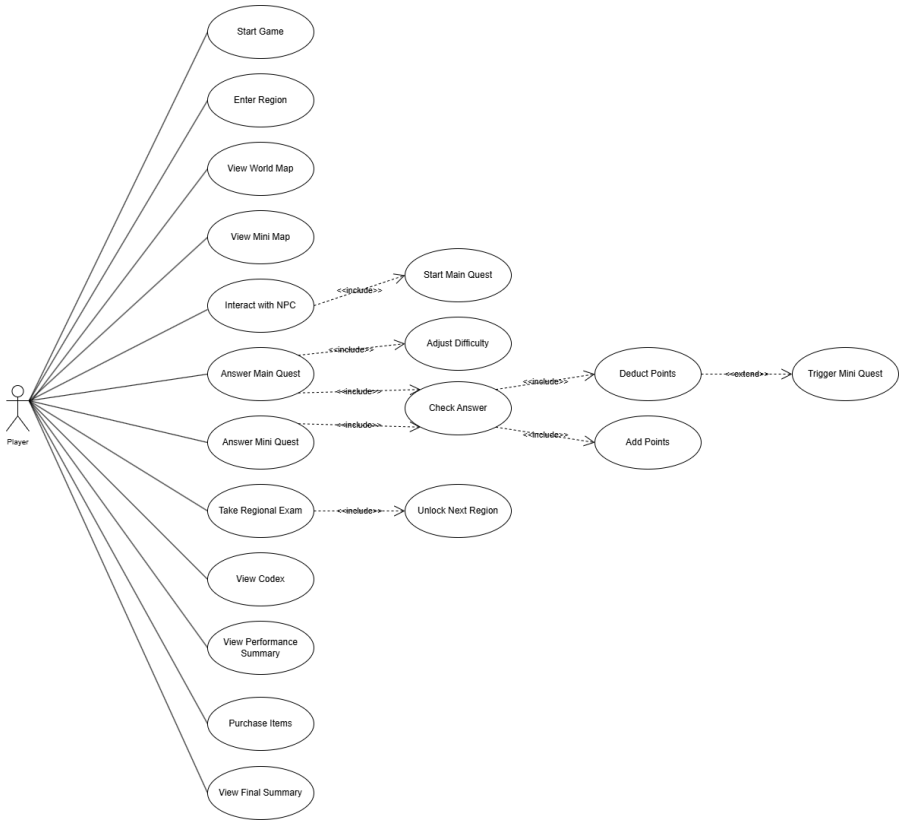


Figure X. Use Case Diagram

Figure X is the use case diagram of AquaQuest. It highlights the actions and interactions of the Player within the game system. The player can initialize basic session and navigation controls, which include starting the game, entering a region, and viewing the world map or mini-map. For core gameplay, interacting with an NPC automatically includes starting a main quest. When answering a main quest, the system includes adjusting the difficulty and checking the answer; checking the answer further includes either adding points for correct inputs or deducting points for mistakes. Deducting points extends to triggering a mini-quest, which requires the player to perform the separate action of answering the mini-quest to check their answer again. To

progress through the curriculum, taking a regional exam includes unlocking the next region. Finally, players can perform secondary tasks such as viewing the codex, checking their performance summary, purchasing items from the shop, and viewing the final game summary.

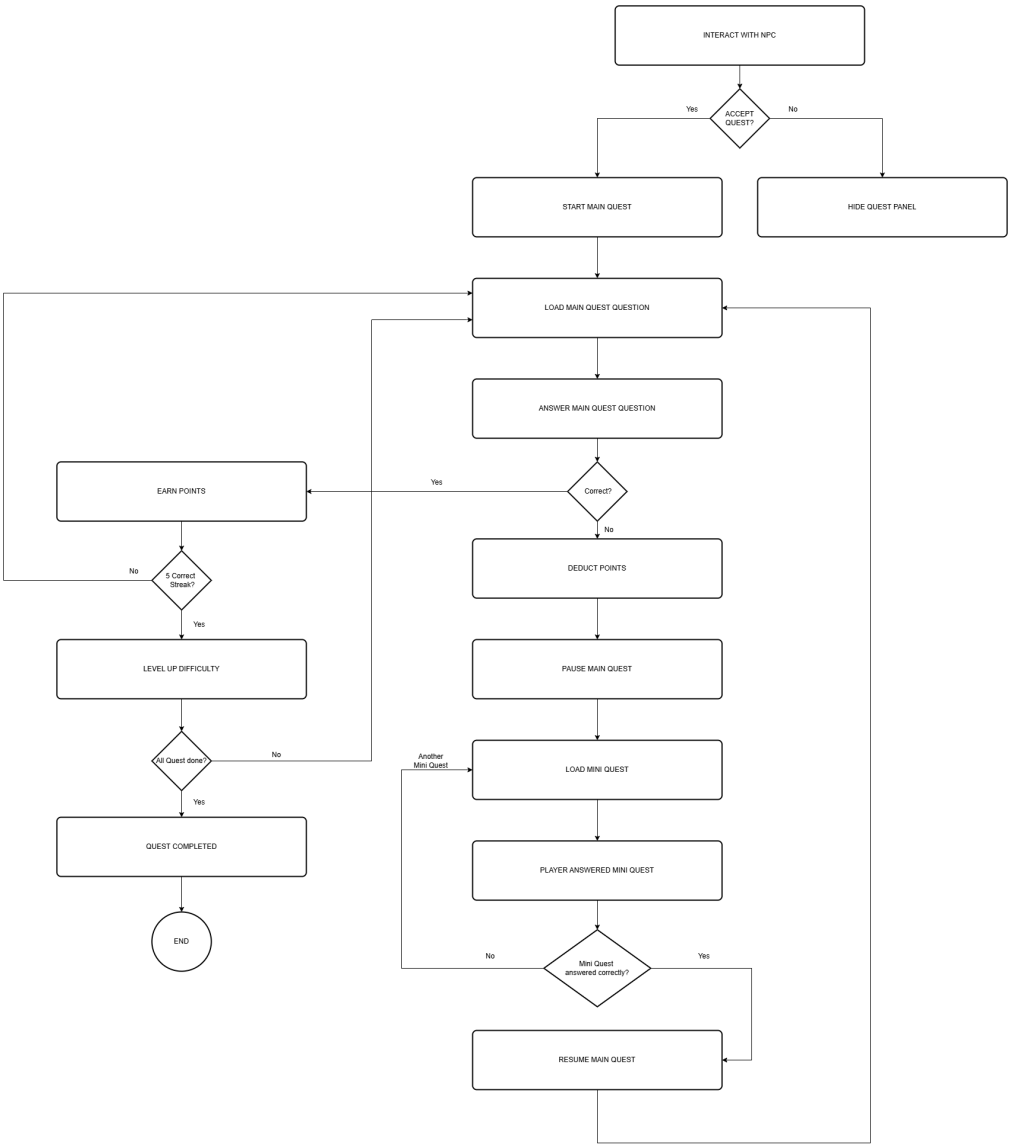


Figure X. Flow Chart

Figure X is the flowchart diagram of AquaQuest. It highlights the game progression, logic gates, and the adaptive learning loops involved in the system. The process begins when the player interacts with an NPC to accept

a quest. When a main quest question is loaded, the player submits their answer. If the answer is correct, the player earns points, and achieving a 5-correct streak will automatically level up the game's difficulty. If all quests are done, the player completes the quest and ends the session. On the other hand, if the player's answer is incorrect, points are deducted, the main quest is paused, and the system loads a mini-quest. The player must answer the mini-quest correctly to resume the main quest; otherwise, they will remain locked in the mini-quest reinforcement loop until mastery is achieved.



Figure X. Spawn Point

When the player enters the game, they will spawn in Region 1. The image shows the first spawn point of the player in Region 1. From this point, players can freely explore, interact with NPCs, and start their first quest. This serves as the home base for all activities in the region.

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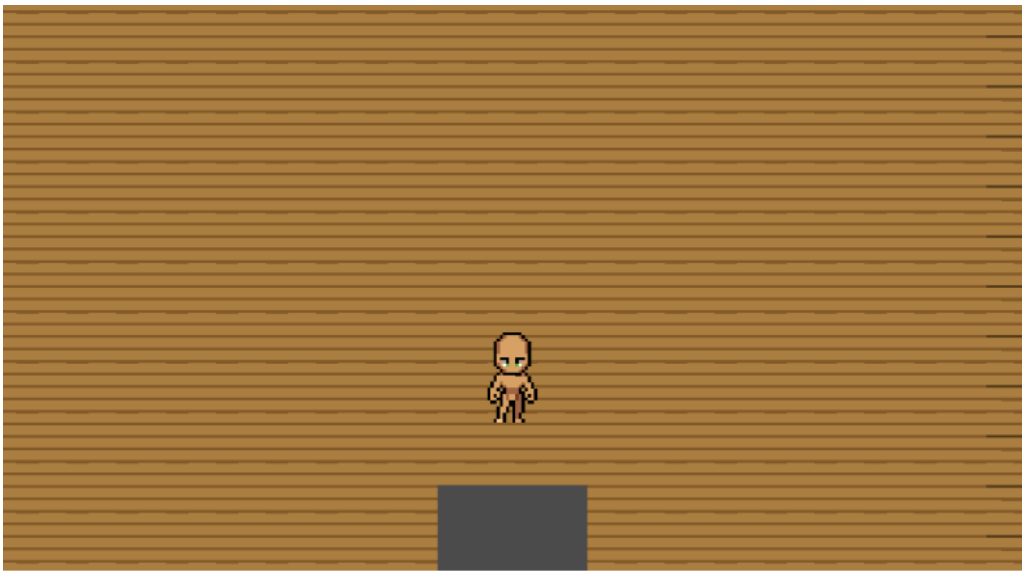


Figure X. Collision

In this image, when the player interacts with the door of a house, they can enter and the scene changes from outside to inside. The players can access the Performance Board, review lessons. This transition mechanic helps players distinguish between exploration areas and interactive learning spaces.



Figure X. Interaction

The player can interact with the NPC to read the dialogue related to the lessons in this region. A dialogue box appears at the bottom of the screen, displaying the NPC's name, portrait, and educational text about Post-Harvest Fisheries topics. This is the primary method for delivering story-driven educational content before quests begin.



Figure X. Quest Notification

The NPC can give a quest to the player, who can either agree to accept it or set it aside for later. A quest notification panel appears showing the title, description and Accept/Decline buttons. If the player declines, the NPC will offer the same quest again after a short delay to encourage engagement. Accepting a quest starts the main learning challenge for that specific lesson.

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Figure X. Main Quest

Once the player accepts the quest, the Main Quest begins and displays questions related to the given topic. The screen shows multiple-choice questions. A progress bar indicates how many questions remain in the current quest. Points are added for correct answers and deducted for incorrect ones, encouraging careful thinking.



Figure X. Mini Quest

If the player answers incorrectly, the Mini Quest system triggers and a new window appears with a simplified review question. After the player answers the Mini Quest correctly, it will resume the Main Quest and restore half of the points that were deducted. This reinforcement system ensures that players understand the concept before moving forward. It turns mistakes into learning opportunities rather than pure penalties.



Figure X. Quest Done

After completing a quest from a specific NPC, the "Quest Done" notification will appear if the player tries to interact with that NPC again.

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Figure X. Region 1

This image shows the whole map of Region 1, displaying all key locations such as NPC houses, quest givers, the shop, and the examination hall.

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PROPOSAL 2

I. **Title:** An Adaptive Web-Based Repository for Post-Harvest Fisheries Modules Using Markov Chain Analysis for Learning State Predictions

Republic of the Philippines
CAVITE STATE UNIVERSITY NAIC
(Formerly CAVITE COLLEGE OF FISHERIES)
Bucana Malaki, Naic, Cavite
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Covered Sustainable Development Goals:

- SDG 14: Life Below Water
- SDG 4: Quality Education
- SDG 2: Zero Hunger

II. Introduction

Modern digital educational repositories and web-based automated learning systems have emerged as powerful paradigms for technical vocational education, transforming static text-based instructional references into interactive, data-driven web environments. However, traditional online repository software platforms suffer heavily from rigid progression mechanics, forcing all learners through a uniform pacing model regardless of their unique prior knowledge, individual cognitive capability, or actual retention rates. To mitigate this constraint, modern computer science frameworks increasingly deploy predictive state architectures. Driven by real-time behavioral telemetry and web-server log mining, these algorithmic engines treat student understanding as a fluid, continuous set of state variables, tracking user navigation histories to automatically forecast academic trajectories and sustain an optimal alignment within institutional curriculum tracks.

Despite the proven efficacy of adaptive software architectures, a critical research gap persists at the intersection of mathematical prediction modeling and specialized domain instruction, particularly within post-harvest fisheries education. Traditional fisheries training relies extensively on manual, text-dense instructional modules, which frequently result in poor cognitive retention and low user engagement. This lack of engagement directly compounds critical real-world issues, as empirical data indicates that physical fish losses across traditional supply chains reach up to 20.0% to 35.0% of total landed volume due to inefficient

processing, improper handling, and inadequate technical knowledge. While statistical prediction frameworks have been successfully implemented in generic commercial Learning Management Systems (LMS) to optimize broad student retention metrics, there is an absolute absence of specialized software frameworks that map real-time, performance-based mathematical state machines to technical maritime or post-harvest vocational guidelines. Consequently, traditional storage portals fail to adapt to the highly varied learning curves of students tackling complex procedural concepts like quality assessment, cold-chain management, and fish processing chemistry.

To bridge these systemic pedagogical and technological gaps, the proposed system introduces *An Adaptive Web-Based Repository for Post-Harvest Fisheries Modules Using Markov Chain Analysis for Learning State Predictions*. Positioned within the specialized domains of Predictive Learning Analytics, Software Engineering, and Computer-Assisted Instruction (CAI), the proposed web system structures the traditional post-harvest fisheries curriculum into interactive, localized digital modules. The core innovation lies in its application of a first-order discrete-time Markov Chain algorithm. By continuously capturing raw user session logs—specifically tracking answer accuracy markers, error distributions, and submission intervals of the last completed assessment or module review—the underlying web engine dynamically computes transition probability matrices to predict the student's next hidden learning state (e.g., State A: Mastery, State B: Remedial). Based on this algorithmic prediction, the repository intelligently updates the web interface to display predictive pop-ups and targeted links to specific prerequisite modules or reinforcement content, ensuring the learner satisfies course guidelines and avoids academic failure.

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The proposed repository framework directly addresses and supports two United Nations Sustainable Development Goals (SDGs): SDG 4 (Quality Education) and SDG 14 (Life Below Water). By deploying a highly personalized, mathematically driven learning prediction engine, the software democratizes technical vocational training, ensuring inclusive, equitable, and effective lifelong learning opportunities that cater to individual student baseline capabilities. Concurrently, by modernizing post-harvest fisheries instruction, the platform equips future fisheries professionals with highly optimized, statistically verifiable technical competencies. Elevating the mastery of proper fish preservation, handling, and value-addition techniques directly mitigates the catastrophic 20.0% to 40.0% post-harvest loss margin, driving sustainable consumption practices, reducing localized food waste, and encouraging the responsible, productive management of marine resources.

III. Client/User Problem and Research Gap

Absence of Intelligent Institutional Storage Tools for Technical Learning. The department completely lacks specialized web-based repository systems capable of dynamically assessing student progress, forcing instructors to rely on non-adaptive, static electronic documents to distribute technical competencies. This lack of diverse educational technology limits the department's capacity to provide interactive, modern web assets that can adapt to individual cognitive variations.

Inability of Static Lesson Material to Address Complex Procedural Concepts. Traditional digital tools present highly detailed post-harvest workflows—such as the stages of fish spoilage, fish deboning, cold-chain

parameters, and advanced preservation techniques—in dense, non-interactive text arrangements. Because these concepts are heavily sequential and physical, static delivery fails to help students visualize structural and biological transformations in real-time, creating severe cognitive bottlenecks.

Cumulative Learning Bottlenecks Due to Unmonitored Pacing Variations. Slower learners experience a compounding misunderstanding of prerequisite topics when navigating traditional linear platforms. Concurrently, instructors lack an integrated web-based evaluation tool to track real-time student capabilities, preventing them from identifying which specific post-harvest module guidelines are causing massive academic failure trends.

Technological Research Gap in Mapping Markov Chain Frameworks to Specialized Domains. While computer science literature frequently analyzes learning behavior using machine learning models, there is a total absence of empirical models that apply runtime Markov Chain transition matrices directly to specialized post-harvest web repositories. Current systems lack an algorithmic infrastructure that can compute live probability values based on a student's immediate historical action and dynamically forecast if the learner will collapse into a remedial tier or advance to a mastery tier.

IV. Objective of the Study

The objective of this study is to design, develop, and implement an adaptive web-based educational repository focused on Post-Harvest Fisheries modules for Fishery and Aquatic Science students. The system aims to provide an intelligent learning environment that integrates core course requirements with a predictive mathematical algorithm to enhance student progression, prevent academic failure, and personalize material navigation.

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The study specifically aims to:

1. Gather user requirements, curate standard post-harvest fisheries course content, and design a scalable web architecture.

2. Incorporate a first-order Markov Chain Algorithm to dynamically predict student learning states based on real-time activity tracking logs, mapping transition matrix paths between Mastery and Remedial thresholds.

3. Develop the web-based system with the following integrated features:

Module Repository Module Structurally hosts, indexes, and presents targeted post-harvest instruction guidelines (e.g., quality assessment, processing chemistry, and cold-chain management).

User Interaction Telemetry Engine: Logs student interaction behaviors, tracking final action accuracy, execution velocities, and mistake counts.

Markov Chain Prediction Processor: Computes live state-to-state probability matrices using the last recorded student log to predict if the user will maintain a mastery baseline or collapse into a remedial state.

Adaptive Recommendation Banner: Automatically displays predictive pop-ups and links to specific prerequisite modules or reinforcement reviews calculated as the highest-probability path required to pass the course.

Learning State Dashboard: Provides students and instructors with a visual breakdown of current transition trends, predicted risk percentages, and curriculum completion metrics.

Integrated Assessment Engine: Executes milestone evaluation blocks to empirically measure retention prior to final module clearance.

4. Conduct system testing to isolate and resolve operational bugs, errors, and matrix calculation variances.

5. Evaluate the platform using ISO 25010 evaluation standards to verify functional suitability, performance efficiency, and reliability.

6. Formulate a structured implementation strategy for institutional deployment within the department.

V. Highlights of Related Literature Review

A systematic overview of real-world statistics and quantitative data proves the mathematical and operational validity of the proposed repository system:

1. **Efficacy of Markovian Sequence Modeling in Live Learning Environments**: According to Paxinou et al. (2021), tracking sequential student actions using a Markov chain model provides a highly precise dynamic evaluation of student performance. In an empirical study evaluating experimental tracking methodologies, sequential tracking engines proved capable of isolating student knowledge states and accurately forecasting innate behavioral abilities across multi-step execution procedures.

2. **Transition Probability Dynamics in Student Engagement States**: Quantitative research in learning analytics confirms that student behavior transitions exhibit distinct probabilistic boundaries over time. In a longitudinal study using Markov models to evaluate digital interactions, data indicated that students in a low-engagement state at an early time interval (T_1) had a 50.0% probability of remaining stuck in that low-performance tier at a later stage (T_2), while displaying a 50.0% chance of recovery; conversely, high-performing

students exhibited an 83.0% probability of maintaining mastery, with only a 17.0% chance of falling into a lower operational state.

3. **Algorithmic Clustering and Problem-Solving Traces**: Educational data mining utilizes historical log streams to segment student actions into distinct behavioral profiles. Maqsood et al. (2024) developed mixture Markov models to cluster problem-solving traces and predict student engagement states. Their implementation of the Expectation-Maximization algorithm on event logs isolated six unique behavioral states—High Knowledge, Less Knowledge, Fill-knowledge Gap, Knowledge Gap, Learn, and Not Interested—confirming that modeling event transitions directly predicts learning outcomes and prevents task disengagement.

4. **Steady-State Probability and Learning Habit Formulation**: The application of state transition matrices to online learning history yields distinct mathematical profiles of student behavior. Research utilizing Markovian learner models to analyze steady-state probabilities (π_i) discovered that a student's long-term academic score correlates directly with their resource use density (r_i) and steady-state persistence patterns. By establishing a scalar Self-Regulated Learning (SRL) index from these Markov chains, the platform systematically maps student strengths and weaknesses to provide automated instructional interventions.

5. **Dynamic Scaffolding and State Stabilization Targets**: Incorporating adaptive computational models within educational web portals minimizes cognitive fatigue. Software frameworks that replace linear paths with automated closed-loop telemetry systems utilize real-time feature extraction to stabilize player success metrics at a balanced winning baseline of roughly 70.0%. This mathematical tuning sustains learner engagement, generating user satisfaction metrics up to 88.4% higher than rigid, non-adaptive educational software structures.

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VI. Innovation and Novelty

The primary innovation of this study lies in its algorithmic mapping of a first-order Markov Chain transition model directly onto a web-based repository hosting specialized, procedural vocational training modules. While traditional academic repositories function strictly as static storage portals where file download structures are completely fixed and linear, this platform introduces an automated, run-time predictive state machine. The system programmatically translates a student's technical performance metrics—such as interaction velocity, quiz answer accuracy, and error frequencies within post-harvest fisheries evaluations—into discrete mathematical state values.

Furthermore, the system establishes a unique dynamic scaffolding feedback loop directly within an institutional web repository environment. Instead of executing post-hoc data mining operations that are only accessible to instructors weeks after an exam, the proposed web platform executes run-time matrix calculations. The underlying engine processes the user's immediate historical action against pre-calculated transition matrices to dynamically project whether the student will maintain academic baseline metrics or drop into a remedial state. If a high failure probability is calculated, the system intelligently adapts the interface, dynamically generating an overlay banner that injects targeted review files and essential prerequisite modules to guide the student back toward a verified mastery path. This approach effectively shifts the application of predictive analytics away from broad, generic course auditing and redirects it toward live, adaptive cognitive mapping within localized maritime and agricultural instructional workflows.

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VII. Target User/Client

The primary target users and direct clients of the proposed web application are the undergraduate students officially enrolled in the Fisheries and Aquatic Science Department of Cavite State University - Naic (CvSU - Naic), located in Bucana Malaki, Naic, Cavite, specifically those undertaking the specialized Post-Harvest Fisheries curriculum.

The secondary clients and key administrative users of the system are the faculty members, instructors, and academic researchers of the CvSU - Naic Fisheries and Aquatic Science Department. Faculty members will utilize the platform's backend administrative engine as an institutional computer-assisted instruction (CAI) tool to cleanly distribute structured post-harvest guidelines, monitor student progression trends, and audit automatically generated learning state dashboards to deliver data-driven pedagogical interventions.

VIII. Significance of the Study

The implementation of the proposed adaptive repository system will yield significant benefits for various stakeholders by modernizing traditional instruction methods through predictive computing technology:

To the Students of the Fisheries and Aquatic Science Department: The platform provides an individualized, automated learning experience that continuously aligns with their unique cognitive pacing. By receiving predictive real-time recommendations, students can easily identify and resolve personal

comprehension gaps in complex post-harvest procedures before facing terminal exams, reducing academic anxiety and increasing overall course success rates.

To the Faculty and Instructors of the Department: The software functions as a highly reliable, automated computer-assisted instruction (CAI) reference tool that eliminates the operational burden of manual performance tracking. The automated learning state dashboard provides instructors with immediate visibility into aggregate classroom difficulty trends, allowing them to rapidly adjust lecture schedules to address specific bottleneck topics.

To the Cavite State University - Naic Administration and Department: The study enhances institutional capabilities by integrating advanced, predictive algorithmic data structures directly into its specialized agricultural and fisheries offerings. This establishes a innovative, digitally archived framework for curriculum delivery that elevates the university's technical profile.

To the Post-Harvest Fisheries Sector: The platform directly supports industry sustainability by training technically competent professionals. Equipping graduates with an optimized, statistically verified mastery of fish preservation and handling helps mitigate the catastrophic 20.0% to 40.0% post-harvest loss margin, supporting local food security and sustainable resource management.

To Future Researchers: This study provides a highly concrete software engineering design pattern and comprehensive empirical documentation on deploying Markov Chain analysis within specialized web-based tutoring applications, serving as a foundational reference for adaptive system engineering in niche domains.

IX. Software and Hardware Requirements

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The development, execution, and evaluation of the adaptive repository require specific web-based software environments and database technologies to effectively process real-time user session data, maintain persistent logs, and execute run-time Markov Chain probability matrix equations. Since the researchers will utilize existing, personally owned hardware assets for development and testing, no new hardware acquisitions are required for this study.

Software Requirements:

Programming Languages & Frameworks (Python / PHP / JavaScript):Python (with Flask or Django frameworks) or PHP will be utilized to construct the robust backend routing, manage user authentication, and execute the core mathematical logic required to calculate state transition matrices. Native JavaScript combined with modern CSS frameworks (such as Bootstrap) will handle frontend interface responsiveness and dynamic banner generation.

Database Management System (MySQL / MariaDB): A relational database system used to store categorized metadata for post-harvest modules, maintain persistent student records, and systematically archive sequential telemetry logs (such as timestamps, mistake logs, and quiz accuracy inputs) required by the prediction engine.

Web Server Environment (Apache / Nginx / XAMPP): XAMPP will serve as the local cross-platform development environment to host the Apache web server and MySQL database components, ensuring rapid prototyping and local network testing before final system deployment.

Web Browser (Google Chrome / Mozilla Firefox): Used as the primary compilation and runtime execution interface to verify frontend compatibility, test

JavaScript event listeners, and ensure seamless delivery of the web application across various client-side devices.

Operating System (Windows 10/11 / Linux): The primary development environments utilized by the researchers to compile scripts, manage database connections, and run server instances.

Hardware Requirements (Personally Owned Assets):

Development Laptop/PC: Equipped with at least a Core i5 or AMD Ryzen 5 processor and a minimum of 8GB RAM to run local web server instances, manage database queries, and operate integrated development environments (IDEs) efficiently without system lag.

X. Source of Information for Data Gathering

To establish a systematic, high-credibility foundation for the proposed software system, the researchers will rely entirely on academic literature, peer-reviewed journals, and official institutional documents.

The primary platform and tool for data gathering is **Google Scholar**, which functions as a verified secondary source repository. Through Google Scholar, the researchers will systematically extract, analyze, and evaluate peer-reviewed research papers, software engineering models, data analytics literature, and mathematical journals focusing on discrete-time Markov Chain applications, learning analytics web log mining, and institutional instructional frameworks. This secondary data will be used to establish the baseline transition probability values, refine the system's predictive mathematical equations, and cross-validate the technical guidelines of post-harvest fisheries pedagogy.

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PROPOSAL 3

I. **Title:** A Mobile-Based Pet Monitoring and Scheduling System Using Decision Tree Algorithm and Location-Based Service Mapping

Covered Sustainable Development Goals:

- SDG 15: Life on Land

  SDG 9: Industry, Innovation, and Infrastructure
- SDG 3: Good Health and Well-being

II. Introduction

Mobile-based health monitoring systems and intelligent scheduling applications have emerged as increasingly practical solutions for managing the care needs of living organisms, extending their utility from clinical human medicine into the broader domain of domestic animal welfare. Among the computing paradigms driving this shift is the Decision Tree algorithm, a supervised machine learning model widely adopted for its interpretability and accuracy in generating rule-based recommendations from structured profile data. Alongside this, Location-Based Service (LBS) mapping has become a standard feature in mobile health platforms, enabling users to discover and navigate to nearby service providers in real time. However, despite the growing adoption of these technologies in human health applications, their integration into a unified mobile platform specifically designed for domestic pet care management remains largely unexplored, particularly within the Philippine context.

This gap is significant given the rapid rise of pet ownership in the country. A 2023 survey by Social Weather Stations (SWS) reported that 64% of Filipino households own at least one pet, with dogs and cats being the most common at 78% and 50% respectively (SWS, 2023). Google search data further reflects this

trend, with pet-related searches reaching over 7.7 million in 2023, representing a 30% growth from 2021 (Vero, 2024). Despite this surge, the supporting infrastructure for pet health management has not kept pace — veterinary access in the Philippines remains critically limited, with only one veterinarian available for every 10,000 pet owners (GlobalPETS, 2025). Most owners continue to depend on informal methods such as handwritten notes or memory to track feeding schedules, medications, and veterinary appointments. Meanwhile, the global mHealth market was valued at $62.73 billion in 2023 and is projected to reach $81.71 billion by 2025 (Emerline, 2025), reflecting a clear global recognition that mobile, algorithm-driven tools are an effective response to health management challenges — a development that the Philippine pet care domain has yet to fully embrace.

To bridge these gaps, the proposed system introduces A Mobile-Based Pet Monitoring and Scheduling System Using Decision Tree Algorithm and Location-Based Service Mapping. The system processes structured pet profile inputs — specifically breed, age, weight, and documented health conditions — through a Decision Tree classification model to generate individualized daily care recommendations for each registered animal. Simultaneously, its integrated LBS mapping module surfaces the real-time locations, operational schedules, and service types of nearby veterinary clinics, grooming establishments, and pet supply stores. By consolidating these two functionalities into a single mobile platform, the system eliminates the fragmented, manual workflows that currently define pet care management in the Philippines and replaces them with an intelligent, personalized, and geographically responsive alternative.

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The proposed system directly addresses two United Nations Sustainable Development Goals (SDGs) (United Nations, 2020). It supports SDG 3 (Good Health and Well-Being) by promoting preventive and consistent animal care, reducing health complications caused by missed routines and delayed veterinary attention. Concurrently, it advances SDG 9 (Industry, Innovation, and Infrastructure) by applying modern mobile computing and machine learning technology to a practical and growing need among Filipino pet owners.

III. Client/User Problem and Research Gap

Most pet owners in the Philippines do not have a proper digital tool to keep track of their pets' daily care needs. Because of this, many are forced to rely on memory or handwritten notes to remember feeding times, medications, and vet schedules, which frequently results in missed routines and neglected care. This problem is made worse for owners who have busy schedules and multiple pets to look after, as manually tracking everything becomes increasingly difficult and unreliable over time.

Existing pet care applications fail to provide care recommendations that consider a pet's specific breed, age, weight, or health condition. Apps like PetDesk and 11Pets only offer basic reminders and generic scheduling features, which means a small Shih Tzu puppy and a fully grown Labrador are given the same care schedule despite having completely different needs. This absence of personalization prevents pet owners from receiving reliable guidance on how to properly care for their specific animals, putting their pets' health at risk.

Pet owners also have no efficient way of locating nearby veterinary clinics, grooming shops, or pet stores through a dedicated pet care application. When a

pet gets sick or needs grooming, owners are forced to search on separate apps or ask around just to find the nearest available service, wasting time and risking the pet's condition in urgent situations. This difficulty is further worsened by the fact that the Philippines only has one veterinarian for every 10,000 pet owners (GlobalPETS, 2025), making the lack of a quick and reliable location tool even more problematic.

No existing pet care application is capable of bringing together pet profile management, personalized care scheduling, automated reminders, and nearby service mapping into one platform. Because these features are unavailable in a single system, pet owners are forced to switch between multiple disconnected apps just to manage their pets' needs, which is inconvenient and causes important care tasks to be frequently forgotten or ignored.

Despite the growing use of machine learning in health-related applications, no existing study in the Philippines has explored the use of a Decision Tree algorithm specifically for generating personalized pet care recommendations. The same is true for integrating real-time location mapping for nearby pet services within a single pet care mobile app, leaving this need completely unaddressed both locally and in existing literature.

IV. Objective of the Study

The general objective of this study is to design, develop, and implement A Mobile-Based Pet Monitoring and Scheduling System Using Decision Tree Algorithm and Location-Based Service Mapping by gathering user requirements, integrating key functionalities, conducting thorough testing, evaluating system performance, and preparing a comprehensive deployment plan.

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Despite the growing use of mobile applications in everyday life, pet owners in the Philippines still lack a dedicated digital tool that can provide care recommendations tailored to their pet's specific profile. Existing applications such as PetDesk and 11Pets only offer basic reminders and generic scheduling features, leaving a clear gap in personalized, algorithm-driven pet care management. Moreover, no existing platform has successfully combined pet profile management, activity logging, automated reminders, and real-time location-based service mapping into a single mobile application, a need that remains unaddressed both locally and in existing literature.

The study specifically aims to:

1. Gather user requirements and design the system architecture of the mobile-based pet monitoring and scheduling system.

2. Integrate the Decision Tree algorithm into the system to process structured pet profile inputs — specifically breed, age, weight, and documented health conditions — to generate individualized daily care recommendations for each registered pet.

3. Develop the system with the following features:

    a. **Account Registration and Pet Profile Management.** Allows pet owners to register an account and create detailed profiles for each of their registered animals.

    b. **Pet Activity Logging.** Enables users to record and monitor daily pet activities, including feeding schedules, exercise routines, and medication intake.

c. **Real-Time Notifications and Reminders.** Delivers automated alerts that notify pet owners of scheduled care routines, pending medications, and upcoming veterinary appointments.

d. **Decision Tree-Based Care Recommendation Module.** Processes each pet's structured profile data to produce personalized care recommendations based on the specific biological and health needs of the animal.

e. **Location-Based Service Mapping.** Displays the real-time locations, operational schedules, and service types of nearby veterinary clinics, grooming establishments, and pet supply stores.

4. Conduct system testing to identify and resolve bugs and errors present within the system.

5. Evaluate the developed system using the ISO/IEC 25010 software quality evaluation standards, assessing its overall performance while identifying areas for further improvement.

6. Develop a comprehensive implementation plan for the deployment of the mobile application.

V. Highlights of Related Literature Review

According to Lyu, Xu, Yang, and Liu (2023), the Decision Tree (DT) algorithm is a supervised machine learning classification method that processes

structured input attributes through a hierarchical series of binary decision nodes to generate rule-based predictive outputs applicable to health decision-making systems. The study aimed to evaluate and compare the predictive performance of three machine learning algorithms— Decision Tree (DT), K-Nearest Neighbor (KNN), and Support Vector Machine (SVM)— in classifying patient medical decision tendencies based on structured health profile inputs. The researchers found that the Decision Tree algorithm achieved a classification accuracy of 80.0% for predicting treatment effect preferences, an F1-score of 0.80, and demonstrated transparent rule generation that could be directly traced to specific input attributes such as medical history, current symptoms, and health condition categories. The study also found that while SVM achieved higher accuracy at 82.0% for treatment effect classification, the Decision Tree model outperformed competing algorithms in interpretability and practical deployment speed. Furthermore, the researchers concluded that the Decision Tree algorithm offers reliable, rule-based classification outputs that are directly traceable to structured input variables, confirming its suitability as a primary recommendation engine in profile-driven health management systems where interpretability and accuracy must coexist (Lyu et al., 2023).

According to an improved ID3 algorithm study (2022), a Decision Tree classification model built on the ID3 algorithm is a machine learning framework that uses information entropy to select the most informative attributes at each branching node, producing a rule-based structure that maps structured profile inputs to precise recommendation outputs. The study aimed to construct an improved ID3-based decision tree model capable of performing physical health evaluation and personalized activity recommendation for college students based

on structured health profile data including physiological measurements and documented health conditions. The researchers found that the improved ID3 algorithm achieved a health decision accuracy of 90.02%, a motion mode recommendation accuracy of 92.58%, and an information entropy ratio of 89.5%, while recording an operation information loss rate of only 4.136% — all significantly outperforming the traditional ID3 algorithm across each benchmark. The study also revealed that the improved model achieved a 12.7% relative increase in processing efficiency and demonstrated greater stability in information transmission when handling structured multi-attribute health profile inputs. Furthermore, the researchers concluded that Decision Tree models based on the improved ID3 algorithm are highly effective for generating personalized care recommendations from structured biological and health condition inputs, establishing a strong empirical basis for integrating tree-based classifiers into mobile health management systems that must produce reliable, attribute-specific care guidance for individual users (Ye et al., 2022).

According to Ofeh and Shadrach (2024), the Decision Tree algorithm is a supervised machine learning technique widely applied in healthcare monitoring systems due to its ability to process structured multi-attribute data and generate interpretable classification rules that directly reflect the relationships between input variables and health outcomes. The study aimed to conduct a comprehensive review of decision tree algorithm applications across healthcare domains, with a focus on evaluating their accuracy, transparency, and practical deployment in health monitoring and diagnostic recommendation systems. The researchers found that decision tree models consistently achieved high classification performance across reviewed healthcare applications, with accuracy rates ranging from 85.0% to 97.0% across diverse health classification tasks, while maintaining

interpretable decision pathways that allowed clinicians and system developers to trace each recommendation directly to the input attributes that triggered it. The study also showed that decision trees demonstrated strong versatility in processing structured health profile data — including age, weight, physiological measurements, and documented health conditions — making them particularly effective for generating individualized care guidance in both human and animal health monitoring contexts. Furthermore, the researchers concluded that the Decision Tree algorithm remains one of the most reliable and interpretable machine learning models for health monitoring applications, and that its combination of high classification accuracy, structured attribute processing, and rule transparency makes it a compelling choice for developers building profile-driven recommendation modules in mobile health management platforms (Ofeh & Shadrach, 2024).

According to Blancaflor, Dela Cruz, Lumbang, and Dalanon (2023), PawHub is a web-based telehealth application that digitally connects companion-animal veterinarians and pet owners in the Philippines through integrated electronic health records, real-time virtual consultation with audio, video, and text messaging features, and electronic prescription capabilities. The study aimed to address the growing operational challenges faced by veterinary clinics in managing the rapidly expanding volume of manually stored pet medical records while simultaneously improving the accessibility of veterinary consultation services for pet owners in the City of San Pedro, Laguna, particularly during and after the disruptions caused by the COVID-19 pandemic. The researchers found that digitizing pet health records through the PawHub platform and enabling remote veterinary consultations significantly reduced the administrative burden of manual record management and improved the speed and accessibility of

veterinary services, with veterinary professionals reporting improved time efficiency and streamlined client communication workflows. The study also revealed that pet owners rated the telehealth platform positively for its ability to eliminate physical barriers to veterinary access, demonstrating that digital consultation tools are a viable and accepted alternative to in-person veterinary visits for routine health inquiries and follow-up care. Furthermore, the researchers concluded that technology-driven pet care platforms integrating digital health records and virtual communication channels are highly effective solutions for bridging the accessibility gap between pet owners and veterinary professionals in the Philippine context, establishing a strong precedent for the development of integrated mobile pet health management systems (Blancaflor et al., 2023).

According to Blancaflor, Agbayani, Guevarra, and Mariano (2024), VETGO is a mobile application engineered to enhance the accessibility and convenience of in-home pet medical services and medication delivery for pet owners in the Philippines by consolidating veterinary service booking, medication ordering, and nearby clinic connectivity into a unified mobile platform. The study aimed to address the critical lack of online platforms providing in-home veterinary services and pet medicine delivery in the Philippines, a gap that forced pet owners to rely on fragmented, time-consuming, and often inaccessible methods for obtaining veterinary care and medications for their animals. The researchers found that 80.0% of users in the Philippines were satisfied with the application's capability to deliver pet medicine and veterinary services, while 60.0% of respondents specifically and positively reviewed the convenience of receiving medical assistance directly at their homes, indicating a strong alignment between the platform's features and the actual service needs of Filipino pet owners. The study also revealed that the integration of service booking, medication ordering, and

veterinary connectivity into a single mobile interface significantly reduced the time and effort required for pet owners to access care, with users reporting measurably improved responsiveness to their pets' health needs. Furthermore, the researchers concluded that dedicated mobile applications consolidating veterinary service access, medication delivery, and pet owner communication represent the most effective digital solution for improving pet care accessibility in the Philippines, directly supporting the development of integrated platforms that eliminate the fragmented, manual workflows currently defining pet health management in the local context (Blancaflor et al., 2024).

VI. Innovation and Novelty

The main innovation of this system is the use of a Decision Tree algorithm that generates personalized care recommendations for each pet based on their specific breed, age, weight, and health conditions. Unlike existing apps like PetDesk and 11Pets which only offer the same generic reminders for all pets regardless of their differences, the proposed system produces care schedules that are uniquely fitted to each animal's biological profile. This makes the system more accurate and practical compared to what is currently available, since a small Shih Tzu puppy and a fully grown Labrador clearly do not have the same care requirements.

Another key feature of the system is its Location-Based Service (LBS) mapping module, which allows pet owners to view the real-time locations, operating hours, and services of nearby veterinary clinics, grooming shops, and pet supply stores directly within the app. No existing pet care application in the Philippines currently offers this feature alongside care scheduling and profile

management in one platform, making this a significant practical improvement over what pet owners currently have access to.

What makes this system truly distinct is that it combines pet profile management, activity logging, automated reminders, algorithm-based recommendations, and location mapping into a single mobile platform — something no existing local or international study has done. This study contributes to the body of knowledge by demonstrating how a Decision Tree algorithm can be applied specifically to pet care recommendation in a mobile context, providing a foundation for future researchers who wish to expand on intelligent and location-aware pet health management systems.

VII. Target User/Client

The primary target users of the proposed system are pet owners residing in Naic, Cavite, regardless of age or gender, who own at least one domestic pet such as a dog or cat and are looking for a more convenient and organized way to manage their pets' daily care needs, health schedules, and veterinary appointments.

Additionally, veterinary clinics, grooming establishments, and pet supply stores located within Naic, Cavite serve as secondary clients, as the system's Location-Based Service mapping module will display their real-time locations, operating hours, and available services to nearby pet owners who are actively searching for accessible pet care resources within the area.

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VIII. Significance of the Study

The development and implementation of the proposed system will benefit the following stakeholders:

To the Pet Owners of Naic, Cavite, this system will help them keep track of their pets' daily care needs without having to rely on memory or handwritten notes. Since the system uses a Decision Tree algorithm, the care recommendations given will be based on their pet's actual breed, age, weight, and health condition, so the guidance is more fitting compared to generic reminders from other apps. They can also use the location mapping feature to easily find nearby vet clinics, grooming shops, and pet stores, which saves time especially when their pet needs immediate attention.

To the Veterinary Clinics, Grooming Establishments, and Pet Supply Stores in Naic, Cavite, being included in the system's location map gives them better visibility to pet owners in the area who are actively looking for nearby services. This makes it easier for potential clients to find and reach them without having to search on separate platforms.

To Future Researchers, this study can serve as a reference for those who want to explore the use of Decision Tree algorithms in pet care or similar recommendation-based mobile systems. It also adds to the existing body of knowledge by showing how location-based services and personalized recommendations can be combined in one platform, which has not yet been done locally.

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IX. Software and Hardware Requirements

The development, execution, and evaluation of the proposed system necessitate specific software development environments to handle the mobile application functionalities, database management, algorithm integration, and location mapping operations. Since the researchers will utilize existing, personally owned hardware assets for development and testing, no new hardware acquisitions are required for this study,

Visual Studio

Description: Visual Studio is a fully integrated development environment (IDE) developed by Microsoft that provides a comprehensive suite of tools for designing, coding, debugging, and deploying applications across multiple platforms.

Justification: Visual Studio is selected as the primary development environment for this project because of its deep native integration with Xamarin, which allows developers to build, test, and deploy cross-platform mobile applications directly within the same IDE without switching between tools. Its built-in debugger, code completion, and project management features make it the most efficient and reliable environment for developing the proposed mobile application.

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Xamarin

Description: Xamarin is an open-source, cross-platform mobile application development framework developed by Microsoft that allows developers to build native Android and iOS applications using C# and the .NET ecosystem.

Justification: Xamarin is selected as the primary mobile development framework because it enables the development of a single shared codebase that can be deployed across multiple mobile platforms, significantly reducing development time and cost. Xamarin also provides access to native device APIs — including GPS and location services required for the LBS mapping module — making it well-suited for building the full feature set of the proposed system.

C#

Description: C# is a modern, object-oriented programming language developed by Microsoft that serves as the primary language of the .NET ecosystem and Xamarin mobile development.

Justification: C# is chosen as the programming language for this system because of its strong compatibility with Xamarin and Visual Studio, and its robust support for implementing algorithmic logic such as the Decision Tree classification module. C# provides a structured and type-safe programming environment that ensures the reliability and maintainability of the system's core recommendation and scheduling functionalities.

MySQL

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Description: MySQL is an open-source relational database management system (RDBMS) that organizes and stores data in structured tables and supports querying through Structured Query Language (SQL).

Justification: MySQL is selected as the database management system for this project because of its reliability, scalability, and widespread use in mobile and web application backends. MySQL will be used to store and manage all structured data within the system, including pet profiles, care schedules, activity logs, user accounts, and nearby service provider information required by the LBS mapping module.

Google Maps API

Description: Google Maps API is a web-based mapping service developed by Google that provides developers with real-time geolocation data, interactive map rendering, distance calculation, and nearby place search capabilities through a programmable interface.

Justification: Google Maps API is chosen as the location service provider for this system because of its accuracy, extensive coverage across the Philippines, and its well-documented integration support for mobile platforms. The Google Maps API will power the system's Location-Based Service mapping module, enabling pet owners to view the real-time locations, operating hours, and service types of nearby veterinary clinics, grooming establishments, and pet supply stores directly within the application.

Initial Budgetary Requirements:

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The study will utilize existing personal computers, open-source tools, and free API tiers for the conduct of the system. Therefore, its initial budgetary requirement has a value of zero.

X. Source of Information for Data Gathering

To establish a systematic baseline for the proposed system, the researchers will conduct structured survey questionnaires and system evaluation sessions with pet owners residing in Naic, Cavite. This process will allow the team to gather user requirements, validate the system's features against the actual care management needs of the target users, and collect evaluation data using the ISO/IEC 25010 software quality standard.

Veterinary references, published journals, and peer-reviewed studies will also serve as secondary sources of information, particularly for constructing the Decision Tree algorithm's training dataset. Breed-specific, age-based, and health condition-driven care guidelines sourced from credible veterinary literature will be used to define the classification rules and recommendation outputs of the system's care recommendation module, ensuring that the generated care schedules are accurate, evidence-based, and appropriate for each registered pet profile.

Electronic Platform

Google Scholar: Used as the primary academic repository to search, evaluate, and extract peer-reviewed literature regarding Decision Tree algorithms, mobile health applications, location-based services, and pet care management systems relevant to the development of the proposed system.

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